INCH-POUND

MIL-C-24643/47B
22 November 1994
SUPERSEDING
MIL-C-24643/47A
14 March 1994

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, TYPE LS1SMWU
(INCLUDING VARIATION LS1SMWA)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-24643.

This cable can be manufactured in two variations: LS1SMWU (unarmored), and LS1SMWA (armored).

REQUIREMENTS:

Qualification required.

Construction (watertight)

First - Copper conductor, tin coated, ASTM B 286, size 22-7. maximum diameter 0.033 inch.
Second - Separator may be used at manufacture's option where required to provide free-stripping insulation.
Third - Cross-linked polyethylene insulation. Nominal wall thickness 0.030 inch; wall thickness may vary from nominal as necessary, in order for completed cable to meet the specified electrical requirements.
* Fourth - Braided shield of AWG No. 34 or No. 36 tin coated copper. Braid angle of 30 to 35 degrees; minimum coverage of 85 percent.
Fifth - Shield insulation of two polyester tapes, type G in accordance with MIL-I-631, outer tape sealed. An alternate shield insulation may consist of one tape, type G of MIL-I-631, plus extruded jacket of a suitable transparent material, minimum average thickness of the transparent material 0.003 inch. The standard identification code shall be applied by method 2 on the inner tape.

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Construction (watertight) (continued)

Construction (continued)

Sixth - The required number of singles (see table I) cabled together with a lay not greater than 24 times the pitch diameter of the layer. Cabling sequence shall be consecutive, starting with no. 1, from the center outward. Fillers shall be used as necessary to make a firm, well rounded assembly.

Seventh - Binder tape, applied helically with overlap.

Construction variant for type LS1SMWU

Eighth - Cross-linked polyolefin jacket. Jacket thickness, 0.50 inch minimum average.

Construction variant for type LS1SMWA

Eighth - Cross-linked polyolefin jacket. Jacket thickness, 0.050 inch minimum average.
Ninth - Braided metal armor

TABLE I. Details.

Military 1/ part no. M24643/47	Size 2/	Number of singles	Overall diameter (inches)			
			TYPE LS1SMWU		TYPE LS1SMWA	
			min	max	min	max
-01UN	70	70	1.465	1.555	1.515	1.605

1/ Unarmored. When armor option is required, see MIL-C-24643 for configuration of part number.
2/ Type LS1SMWU and LS1SMWA.

EXAMINATION AND TESTS:

Basic electrical:	Requirements
Conductor Resistance - ohms/1000 feet at 25°C, maximum	17.71
Voltage withstand - volts, root mean square, minimum	
Conductor to shield	5000
Shield to shield	500
Insulation resistance - Megohms/1000 feet minimum.	
Conductor to shield	500
Shield to shield	100
* Conductor and shield continuity	No failure

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EXAMINATION AND TESTS (continued):	Requirements
Group A:	
Visual and dimensional	No failure
Watertightness - see MIL-C-24643 for limits of water leakage	No failure
Capacitance - 1 kilohertz (KHz), pF/ft, maximum	35
Attenuation - at 400 MHz, decibel per 100 ft, maximum	18
Group B:	
Cross-linked proof test (percent, maximum)	
Insulation	50
Jacket (When tested at 200°C)	50
Cold bending, cable - at minus 0 ± 2°C, mandrel maximum diameter 20 inches	No damage
Drip - 95 ± 1°C	Zero
Tear - pounds per inch thickness, minimum (ASTM D 470)	35
Physicals (unaged)	
Insulation	
Tensile strength - lb/in^2, minimum	1800
Elongation - percent, minimum	250
Jacket (cable)	
Tensile strength - lb/in^2, minimum	1300
Elongation - percent, minimum	160
Group C:	
Physicals (aged)	
Insulation	
Air oven	
Tensile strength - percent of unaged, minimum	80
Elongation - percent of unaged minimum	80
Jacket (cable)	
Air oven	
Tensile strength - percent of unaged, minimum	60
Elongation - percent of unaged minimum	60
Hot oil immersion	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Shrinkage	No failure
Heat distortion - percent of unaged, maximum	30
* Permanence of printing (jacket) - cycles, minimum	125
Cable sealant removability	No failure
Armor (type LS1SMWA only) - conformance to material, construction and coverage	No failure
Shield - conformance to material, construction and coverage	No failure
Group D:	
Flame propagation (cable)	No failure

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QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

Aging and compatibility (cable) (125± 5°C.)	No failure
Abrasion resistance (jacket) - scrapes, minimum	75
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Shield insulation	2
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Smoke index, maximum	
Jacket	25
Fillers	45
Insulation	45
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure

UNIT ORDERING LENGTHS: 1000 feet nominal

NOTE: The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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